ROBOTICS & AUTOMATION

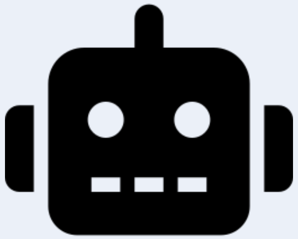
A Simple Guide to Machines That Work For Us

THE BASICS

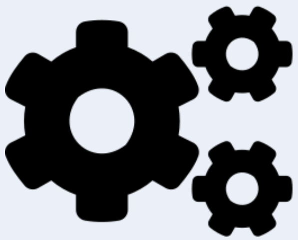
What is Robotics?

Robotics is the science of designing and building machines called **robots** that can sense, think, and act in the physical world.

- Robots can be programmed to repeat tasks, or smart enough to make their own decisions.
- They combine mechanical parts, electronics, and software.



What is Automation?



Automation means using technology to perform tasks **with little or no human help**.

- It can be as simple as a thermostat, or as complex as a factory line.
- Automation doesn't always need a robot — but robots make it more flexible.

THE BASICS

Robotics vs. Automation

Robotics

- The physical machine
- Has a body: arms, wheels, sensors
- Can move and interact with the world

Automation

- The process or system
- Can be software, sensors, or rules
- Focuses on removing manual steps

WHERE IT STARTED

A Short History

1950s

First industrial robot arm built for factory work

1970s

Robots gain sensors and simple computer control

2000s

Robots enter homes, hospitals, and warehouses

Today

AI gives robots the ability to learn and adapt

ANATOMY

Key Parts of a Robot

Sensors

Gather info like light, sound, distance

Controller

The "brain" that processes decisions

Actuators

Motors that create movement

Power Source

Battery or electricity to run it

ANATOMY

How a Robot "Thinks"

Sense

Collects data from its surroundings

Think

Processes data and decides what to do

Act

Moves or performs the task

This loop repeats — often many times per second.

SECTION

Types of Robots

Four common categories, based on where and how they work

Industrial Robots

Used on factory floors to build products quickly and precisely.

- Weld, paint, or assemble parts
- Work 24/7 without breaks
- Common in car and electronics factories

TYPES OF ROBOTS

Service Robots

Help people with everyday tasks in homes and businesses.

- Vacuum cleaners and lawn mowers
- Delivery robots in hotels and offices
- Designed for safety around people

Mobile & Autonomous Robots

Move through spaces on their own using sensors and maps.

- Self-driving cars and delivery drones
- Warehouse robots that carry goods
- Navigate around obstacles automatically

Humanoid Robots

Built to look and move like a human being.

- Walk on two legs, use arms and hands
- Used for research and human interaction
- Still an emerging, developing field

SECTION

Types of Automation

Not all automation works the same way — here are three levels

TYPES OF AUTOMATION

Fixed Automation

Machines built to do one task, the same way, every time.

- Best for very high-volume production
- Hard and costly to change or reprogram
- Example: bottling and packaging lines

TYPES OF AUTOMATION

Programmable Automation

Machines that can be reprogrammed for different tasks.

- Good for medium-volume production
- Switches between product batches
- Example: robotic arms in a car plant

TYPES OF AUTOMATION

Flexible Automation

Smart systems that adjust automatically with little downtime.

- Great for many small, varied tasks
- Uses sensors and software to adapt
- Example: modern smart factories

SECTION

Where We See It Today

Robotics and automation now touch almost every industry

IN PRACTICE

Manufacturing

The birthplace of industrial robotics — still its biggest user.

- Assembles, welds, and packages products
- Improves speed and consistency
- Reduces repetitive strain on workers

IN PRACTICE

Healthcare

Supports doctors and improves patient care.

- Robotic-assisted surgery for precision
- Delivery robots move supplies and meds
- Automation speeds up lab testing

IN PRACTICE

Agriculture

Helps farmers grow more food with less labor.

- Robots plant seeds and pick fruit
- Drones monitor crop health from above
- Automated irrigation saves water

IN PRACTICE

Logistics & Warehousing

Keeps goods moving from factory to front door.

- Robots sort and move packages
- Automated systems track inventory
- Speeds up online order fulfillment

IN PRACTICE

Space Exploration

Goes where it is too dangerous for humans to go — yet.

- Rovers explore the surface of Mars
- Robotic arms service space stations
- Probes travel beyond our solar system

SMARTER MACHINES

The Role of AI in Robotics

Artificial Intelligence gives robots the ability to **learn and adapt**, instead of just following fixed instructions.

- Recognizes images, speech, and objects
- Predicts outcomes and plans next steps
- Improves performance over time from data

WHY IT MATTERS

Benefits of Automation

Speed

Completes tasks faster than manual work

Consistency

Delivers the same quality every time

Safety

Takes on dangerous or repetitive jobs

Cost Savings

Lowers long-term operating costs

THE OTHER SIDE

Challenges & Risks

Job Changes

Some roles shift or disappear

High Setup Cost

Equipment and training are expensive

Skill Gaps

Workers need new training

Safety & Ethics

Needs careful rules and oversight

LOOKING AHEAD

The Future: Working Together

The next wave is **collaborative robots** ("cobots") that work safely alongside people, not instead of them.

- Smarter AI will let robots handle more complex, changing tasks
- Robots and humans will increasingly share the same workspace
- New jobs will focus on designing, training, and maintaining robots

SUMMARY

Key Takeaways

Robotics is the machine

the physical body that senses and acts

Automation is the process

using technology to reduce manual work

AI makes it smarter

helping systems learn and adapt over time

The goal is partnership

machines handling tasks so people can focus elsewhere

Thank You

Questions & Discussion